

Exercise Sheet Generative Models

Exercise 1: Global Optimality of the GAN objective (10 + 10 + 20 P)

In this exercise, we want to show that the global optimal solution for the minimax game

$$\min_G \max_D V(D, G) = \min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{x}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (1)$$

for training Generative Adversarial Networks is that the data distribution gained from sampling from p_g is equal to the real data distribution p_{data} .

(a) Therefore, we first consider the optimal discriminator D for any given generator G . Show that for fixed G , the optimal discriminator D is

$$D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \quad (2)$$

Hint: For any $(a, b) \in \mathbb{R}^2 \setminus \{0, 0\}$, $y \in [0, 1]$, the function $f(y, a, b) = a \log(y) + b \log(1 - y)$ achieves its maximum at $\frac{a}{a+b}$.

$$\begin{aligned} \arg\max_D V(G, D) &= \arg\max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{x}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \\ &= \arg\max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g(\mathbf{x})} [\log(1 - D(\mathbf{x}))] \\ &= \arg\max_D \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) d\mathbf{x} + \int_{\mathbf{x}} p_g(\mathbf{x}) \log(1 - D(\mathbf{x})) d\mathbf{x} \\ &= \arg\max_D \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) + p_g(\mathbf{x}) \log(1 - D(\mathbf{x})) d\mathbf{x} \\ &= \arg\max_D p_{\text{data}} \log(D) + p_g \log(1 - D) \\ &= \arg\max_D f(D, p_{\text{data}}, p_g) \end{aligned}$$

(b) Show that the maximum $C(G) = \max_D V(G, D)$ of the training criterion can be reformulated to:

$$\begin{aligned} C(G) &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[\log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \quad (3) \\ C(G) &= \max_D V(G, D) \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{x}}} [\log(1 - D_G^*(G(\mathbf{z})))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[\log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \end{aligned}$$

(c) Show that the global minimum of $C(G)$ is $C^* = -\log(4)$ and that reaching it is equivalent to $p_g = p_{\text{data}}$.

Hint: Use the fact, that the the Jensen Shannon Divergence $JSD(P\|Q) = \frac{1}{2} (KL(P\|M) + KL(Q\|M))$ is always positive.

$$\begin{aligned}
C(G) &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(\mathbf{x})}{P_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[\log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \\
&= -\log(4) + \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\log \frac{2p_{\text{data}}(\mathbf{x})}{P_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[\log \frac{2p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \\
&= -\log(4) + KL \left(p_{\text{data}} \parallel \frac{p_{\text{data}} + p_g}{2} \right) + KL \left(p_g \parallel \frac{p_{\text{data}} + p_g}{2} \right) \\
&= -\log(4) + 2 \cdot JSD(p_{\text{data}} \parallel p_g)
\end{aligned}$$

Exercise 2: Reformulating the loss function of diffusion models (20 P)

(a) Show that

$$L_{vlb} = \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} \mid \mathbf{x}_0)} \right]$$

can be reformulated to:

$$L_{vlb} = L_0 + L_1 + \dots + L_{T-1} + L_T$$

where

$$\begin{aligned}
L_0 &= -\log p_\theta(x_0 \mid x_1) \\
L_{t-1} &= D_{KL}(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)) \\
L_T &= D_{KL}(q(x_T \mid x_0) \parallel p(x_T))
\end{aligned}$$

with the help of the Markov assumption in Diffusion models.

Substituting $s_1 = -\log p(\mathbf{x}_T) - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}$.

$$\begin{aligned}
L_{vlb} &= \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \log \frac{\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{\prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \\
&= \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \log \frac{p_\theta(\mathbf{x}_0 | \mathbf{x}_1)}{q(\mathbf{x}_1 | \mathbf{x}_0)} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_t | \mathbf{x}_{t-1})} \right] \\
&= \mathbb{E}_q \left[s_1 - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[s_1 - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)} - \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \right] \\
&= \mathbb{E}_q \left[s_2 - \sum_{t=2}^T \log q(\mathbf{x}_{t-1} | \mathbf{x}_0) + \sum_{t=2}^T \log q(\mathbf{x}_t | \mathbf{x}_0) \right] \\
&= \mathbb{E}_q \left[s_2 - \sum_{t=1}^{T-1} \log q(\mathbf{x}_t | \mathbf{x}_0) + \sum_{t=2}^T \log q(\mathbf{x}_t | \mathbf{x}_0) \right] \\
&= \mathbb{E}_q \left[s_2 - \log q(\mathbf{x}_1 | \mathbf{x}_0) - \sum_{t=2}^{T-1} \log q(\mathbf{x}_t | \mathbf{x}_0) + \sum_{t=2}^{T-1} \log q(\mathbf{x}_t | \mathbf{x}_0) + \log q(\mathbf{x}_T | \mathbf{x}_0) \right] \\
&= \mathbb{E}_q [s_2 - \log q(\mathbf{x}_1 | \mathbf{x}_0) + \log q(\mathbf{x}_T | \mathbf{x}_0)] \\
&= \mathbb{E}_q \left[\log p(\mathbf{x}_T) - \log \frac{p_\theta(\mathbf{x}_0 | \mathbf{x}_1)}{q(\mathbf{x}_1 | \mathbf{x}_0)} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)} - \log q(\mathbf{x}_1 | \mathbf{x}_0) + \log q(\mathbf{x}_T | \mathbf{x}_0) \right] \\
&= \mathbb{E}_q \left[-\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T | \mathbf{x}_0)} - \sum_{t=2}^T \log \frac{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)}{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)} - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right] \\
&= D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) + \sum_{t=2}^T D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) \\
&\quad - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)
\end{aligned}$$

Exercise 3: Programming (40 P)

Download the programming files on ISIS and follow the instructions.
Generate MNIST with diffusion model in PyTorch.